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How Well Do Federal Traffic Count Stations Cover Rural Roads? Evidence from the Travel Monitoring Analysis System, 2015–2024

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License: CC BY 4.0. Data and code: https://github.com/Emmanuel-Oti-Nani/tmas-rural-coverage

Key findings

- In 2024, 7,859 permanent count stations reported hourly traffic volume to the Federal Highway Administration's Travel Monitoring Analysis System (TMAS). Of these, 48.9% are on roads coded rural. Rural roads carry 31.1% of vehicle-miles traveled (VMT), make up 70.3% of public road miles, and account for 47.8% of road deaths.
- Per vehicle-mile, rural roads are not under-covered: they have 3.75 stations per billion VMT against 1.77 for urban roads. Per road mile, they have 2.5 times fewer stations (1.30 vs 3.21 per 1,000 miles).
- Coverage is concentrated on rural interstates and arterials. Rural collectors and local roads are 91.1% of rural mileage and carry 34.0% of rural VMT, and 40.5% of rural road deaths in 2020–2024 (39,758 deaths) occurred on them. Yet they have 13.1% of rural stations (504). Rural local roads have 29 stations for 2.03 million miles: 412 deaths per station, against 11 on rural interstates.
- 1,130 of 3,143 counties (36.0%) had no station reporting volume in 2024, including 46.6% of mostly rural counties. Those counties recorded 23.4% of the nation's rural road deaths in 2020–2024.
- From 2015 to 2024 the number of stations in the TMAS station file grew 46% on urban roads and 20% on rural roads. The rural share fell from 51.2% to 46.2%.
- The public TMAS download contains no bicycle or pedestrian count stations, so rural nonmotorized coverage cannot be assessed from public data.

1. Why station coverage matters

States operate permanent, continuously recording traffic counters and report their data to FHWA monthly through TMAS (OMB Control No. 2125-0587). These continuous count stations do two jobs. Directly, they measure traffic where they sit. Indirectly, they supply the seasonal and day-of-week adjustment factors that convert short, temporary counts into annual average daily traffic on the much larger network that has no permanent counter. Annual average daily traffic, in turn, is the basis for the vehicle-

miles traveled estimates behind national travel trends and fatality rates per 100 million VMT.

The Traffic Monitoring Guide (TMG) asks states to build factor groups from continuous stations and recommends "a minimum of 6 per factor group" for non-recreational roads (FHWA, 2022, Section 3.1.4.5). Its minimum recommended volume factor groups place every rural road below the interstate (HPMS functional codes 2 through 7, from rural freeways to rural local roads) in one "Other Rural" group (Table 3-1). A state can therefore meet the minimum with stations on rural arterials only, and apply their seasonal pattern to rural collectors and local roads that carry a different mix of farm, freight, school, and recreational traffic.

FHWA has published a 60-day notice to extend the TMAS collection (FHWA, 2026). This report measures rural coverage with public TMAS data to inform comments on that notice.

2. Data and methods

Stations. FHWA's annual TMAS station files list each station by direction and lane. The 2024 file has 38,886 lane records, which collapse to 10,700 unique stations (state plus station ID). Each record carries a functional system code with a rural (R) or urban (U) suffix and coordinates. Not every listed station reports data, so the main analysis uses the 7,859 stations that appear in at least one of FHWA's twelve 2024 monthly hourly-volume files ("active"). The 2,841 listed stations without 2024 volume records may report classification or weight data only, or may be inactive. The station file's discontinuation field is not consistent enough to tell these cases apart.

Rural and urban. The main analysis uses FHWA's own rural/urban code, which matches the definition used for road miles and VMT in Highway Statistics. As a check, each station was also placed in or out of a 2020 Census urban area. The two labels agree for 87.6% of active stations, and under the Census definition the rural share of active stations is 56.0%.

Denominators. Road miles and VMT by functional system come from Highway Statistics 2024 Tables HM-20 and VM-2 (50 states and DC). Road deaths come from the author's U.S. Rural Road Fatality County Panel (Nani, 2026), which assigns each FARS fatal crash to a county and classifies it as rural or urban using 2020 Census urban areas. Stations were assigned to counties from their coordinates.

Trend. Station counts for 2015–2024 use each year's station file with the same rules. The 2025 file uses a revised record layout with two-digit functional codes and no R/U suffix. It is shown separately and treated as provisional.

3. Results

3.1 National shares

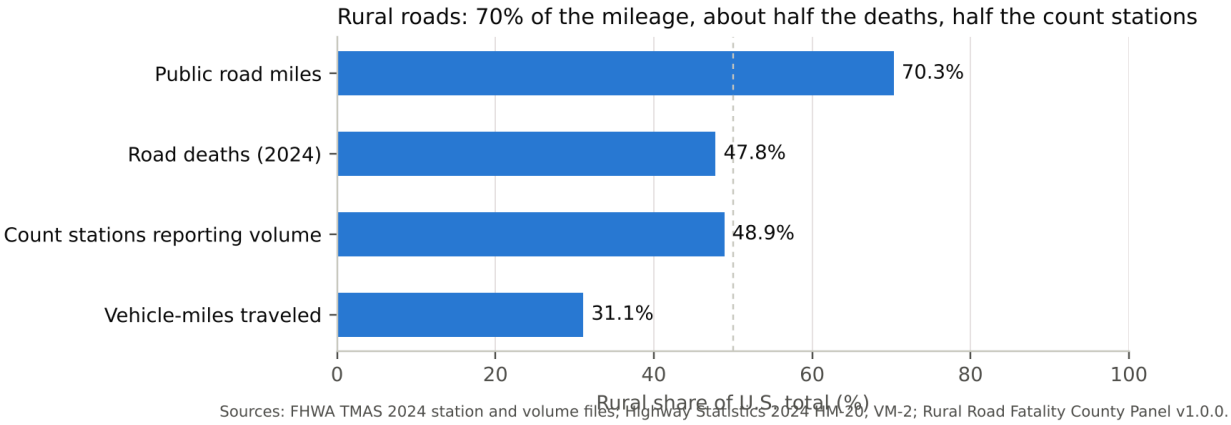


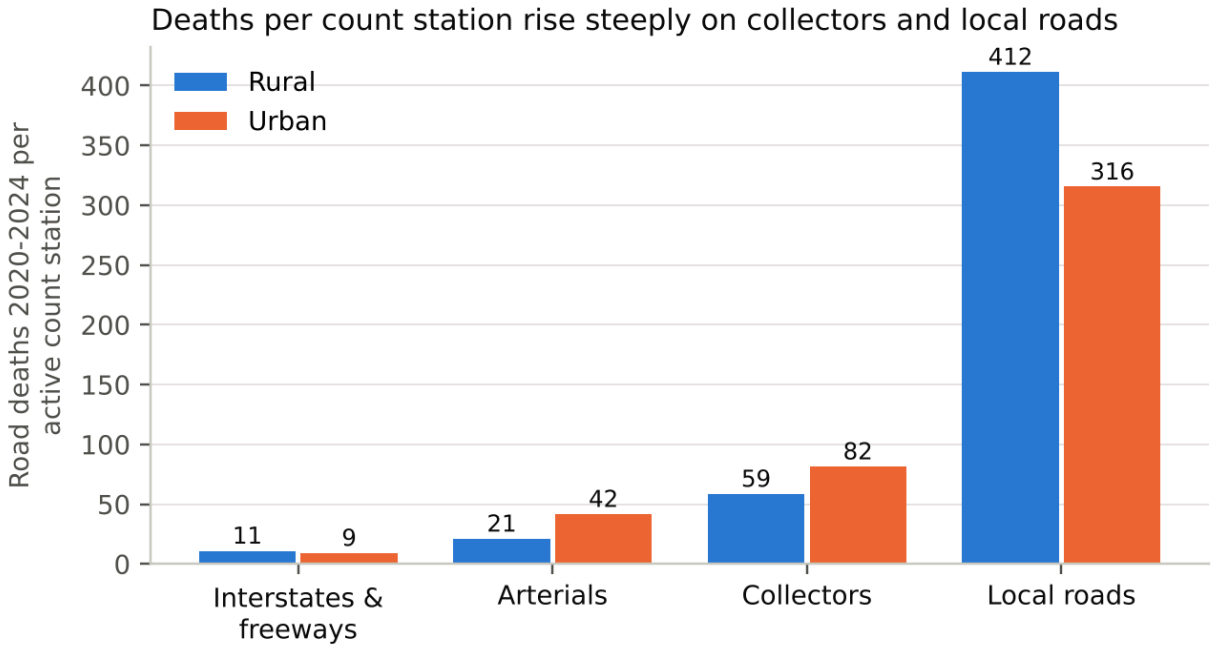
Figure 1. Rural share of U.S. public road miles, VMT, road deaths, and active TMAS count stations, 2024.

Rural stations number 3,846, against 4,013 urban. That is 1.30 stations per 1,000 rural road miles (one per 769 miles) and 3.21 per 1,000 urban miles (one per 312 miles). Relative to traffic, rural roads have more stations than urban roads. The station network is roughly proportional to rural deaths, but far below rural mileage.

3.2 By road class

Area	Road class	Active stations	Road miles	VMT (billion)	Stations per 1,000 mi	Stations per billion VMT	Deaths 2020-24	Deaths per station
Rural	Interstates and other freeways	1,255	36,042	312.5	34.82	4.02	13,791	11.0
Rural	Arterials	2,087	227,711	363.9	9.17	5.73	44,121	21.1
Rural	Collectors	475	667,004	215.5	0.71	2.20	27,824	58.6
Rural	Local roads	29	2,025,860	132.9	0.01	0.22	11,934	411.5
Urban	Interstates and other freeways	2,317	31,657	845.2	73.19	2.74	21,379	9.2
Urban	Arterials	1,521	176,671	885.4	8.61	1.72	63,470	41.7
Urban	Collectors	144	155,034	239.2	0.93	0.60	11,729	81.5
Urban	Local roads	31	888,474	299.4	0.03	0.10	9,779	315.5

Table 1. Active TMAS stations (2024, FHWA rural/urban code) against road miles and VMT (Highway Statistics 2024) and road deaths (2020–2024, Census 2020 urban areas) by road class. Interstates and freeways are HPMS codes 1–2, arterials 3–4, collectors 5–6, local roads 7. Urban deaths are all deaths on the class minus rural deaths.



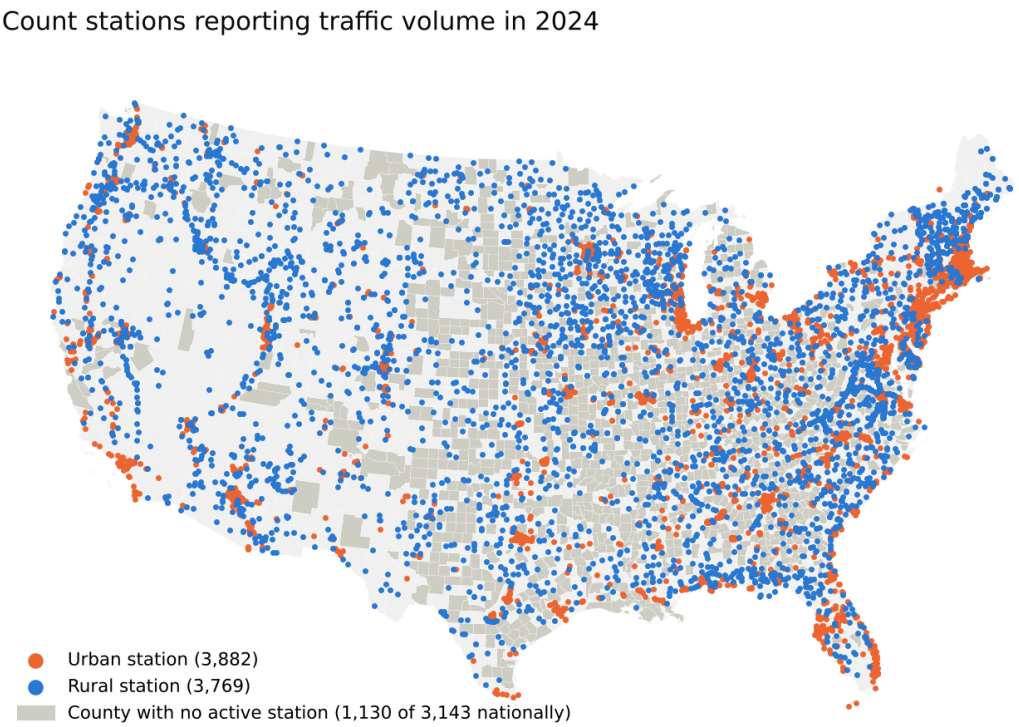
Stations: TMAS 2024, FHWA rural/urban code. Deaths: Rural Road Fatality County Panel (Census 2020 urban areas).

Figure 2. Road deaths in 2020–2024 per active count station, by road class.

The gap is a road-class gap more than a rural-urban gap. Rural interstates have 39.2 stations per 1,000 miles. Rural minor collectors have 0.19, and rural local roads 0.01. Even per vehicle-mile, rural collectors and local roads have 1.45 stations per billion VMT, against 4.94 on rural interstates and arterials. Urban collectors and local roads are covered even more thinly per VMT. In rural areas, however, these roads are where most of the mileage lies and where 40.5% of deaths occur.

By state, 34 of 50 states have at least six active stations on rural collectors and local roads combined, the TMG's minimum for a single factor group. 16 do not, and MI and RI reported none.

3.3 By county



Source: FHWA TMAS 2024 station and monthly volume files; Census 2023 county boundaries. Alaska and Hawaii not shown.

Figure 3. Active TMAS stations in 2024 and counties without one (lower 48 states).

1,130 counties (36.0%) had no active station in 2024. Among the 2,010 counties where at least half the population lived outside urban areas in 2020, 46.6% had none, against 17.0% of other counties. Counties without a station hold 33.6% of rural road miles and recorded 22,963 rural road deaths in 2020–2024 (23.4% of the national rural total). A county without a station still has traffic estimates. They come from short counts factored with patterns measured elsewhere.

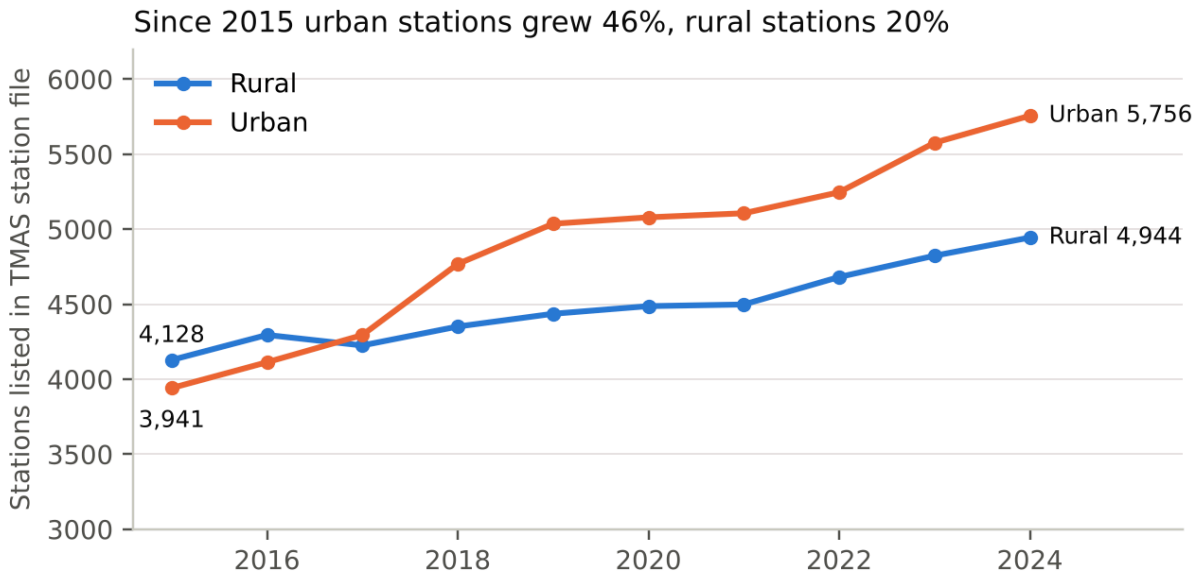
3.4 By state

State	Active rural stations	Rural road miles	Rural stations per 1,000 mi	Rural collector and local stations	Rural share of stations / of deaths (%)
AR	17	80,590	0.21	4	81.0 / 70.9
IL	46	96,345	0.48	1	39.3 / 40.0
OK	48	96,268	0.50	9	60.0 / 66.8
NE	46	86,923	0.53	10	66.7 / 69.0
NM	36	60,427	0.60	3	48.6 / 60.8
KS	75	124,932	0.60	7	78.9 / 67.1
SD	49	78,459	0.62	3	90.7 / 84.0
IN	44	68,769	0.64	12	45.4 / 55.5
LA	28	43,254	0.65	5	53.8 / 50.3
MD	10	13,770	0.73	3	17.5 / 29.8
			…		
ME	105	19,301	5.44	26	77.2 / 81.1
NH	75	11,155	6.72	16	44.6 / 58.0
VA	358	47,554	7.53	14	47.5 / 55.7
DE	26	2,932	8.87	5	28.0 / 37.3
HI	19	1,382	13.75	4	22.9 / 41.2

Table 2. The ten states with the fewest active rural stations per 1,000 rural road miles, and the five with the most (2024). The median state has 1.13. Full table in the data repository.

In 31 of 50 states, the rural share of active stations is below the rural share of road deaths. Missouri, for example, has 102 active rural stations (0.95 per 1,000 rural miles), 11 of them on rural collectors or local roads.

3.5 Trend, 2015–2024



Source: FHWA TMAS station files 2015-2024 (one row per state and station ID; FHWA rural/urban code).

Figure 4. Stations listed in the TMAS station file by FHWA rural/urban code, 2015–2024.

The station file grew by 2,631 stations between 2015 and 2024. 1,815 were urban and 816 rural. The provisional 2025 file lists more stations under the new layout, with a rural share of 43.1%.

3.6 Bicycle and pedestrian counts

TMAS accepts nonmotorized count data, and the TMG nonmotorized station record already includes latitude, longitude, and a functional class and area type (urban or rural) field (FHWA, 2024). FHWA's public TMAS download, however, contains only motorized station records. A 2025 multistate review reported that "only a handful of jurisdictions" had submitted nonmotorized data to TMAS (TETC, 2025, p. 26). The rural coverage of federal bicycle and pedestrian counts therefore cannot be measured from public sources. Rural pedestrian and cyclist deaths have no published exposure denominator from TMAS at all.

4. Limitations

1. "Active" means at least one day of hourly volume in a 2024 monthly file. Stations reporting only classification or weight data are counted as not active.
2. Station and mileage counts use FHWA's rural/urban code (adjusted urban boundaries). Deaths use 2020 Census urban areas. The labels agree for 87.6% of stations. Per-station death ratios mix the two and should be read as orders of magnitude.

3. Station functional codes are as reported by states, and 50 2024 stations lack usable coordinates (excluded from county counts).
4. Station counts measure the network, not the accuracy of the resulting traffic estimates. A sparse network does not by itself prove estimates are biased. It means their accuracy on low-class rural roads cannot be checked from TMAS.
5. Some states give each direction of travel its own station ID. Collapsing active stations of the same road class within about 300 meters into one physical site gives 7,085 sites. The rural share (49.4%) and the share of rural sites on collectors and local roads (13.4%) barely change.
6. States also run short-duration counts that are not reported to TMAS and are not counted here.

5. Implications

The TMAS station network follows traffic volume, which is what it was designed to do. Rural road safety programs, however, need exposure on the rural collectors and local roads where many rural deaths occur. On those roads TMAS has few or no permanent stations, so the denominators of rural fatality rates rest on factors borrowed from busier roads. Low-cost steps within the existing collection could make this visible and start to close the gap. These include publishing coverage by road class each year, reporting a rural low-volume factor group, publishing an active-station flag, and releasing the nonmotorized station file.

Data availability

All inputs are public. Code, processed tables, a quality-assurance report, and provenance records (file sizes and SHA-256 hashes) are in the project repository and archived with this report at DOI 10.5281/zenodo.23772244.

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